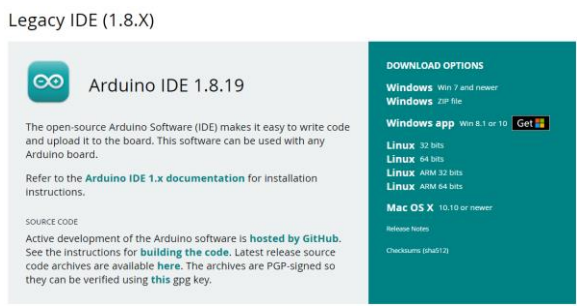
# Getting Start ESP32: ESP32 GPIO + ESP32 Interface

## 1/10. Arduino IDE - 1.8.19 Install + Blink Test

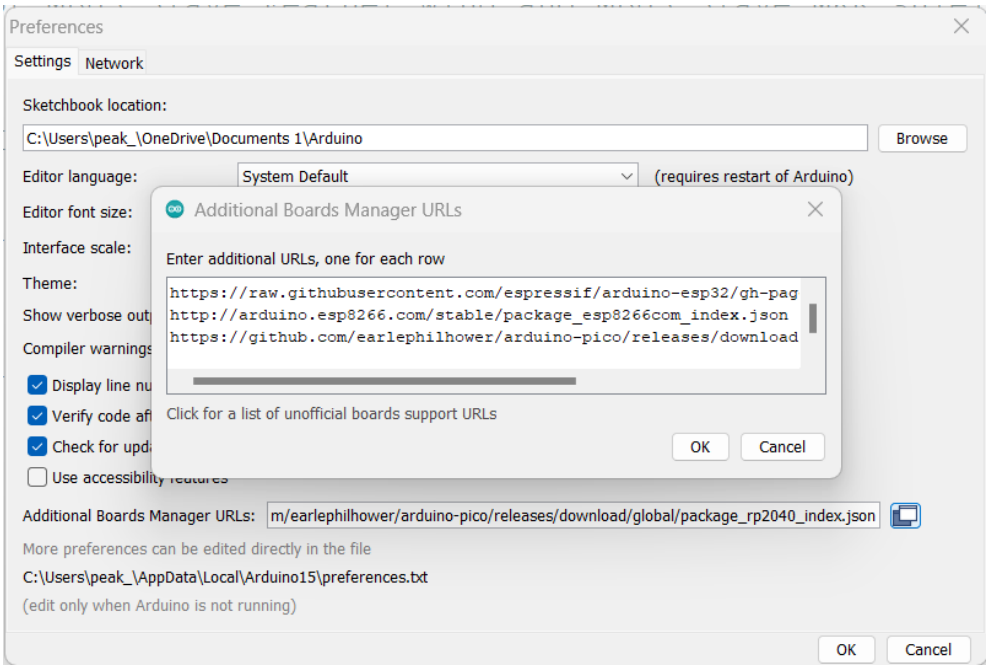
1. Read https://www.cybertice.com/p/5089
2. Read https://support.arduino.cc/hc/en-us/articles/360019833020-Download-and-install-Arduino-IDE
3. Download "arduino-1.8.19-windows.exe" and Install




Windows, Linux or Mac

Just Download

4. Add ESP32 Board URLs: File → Preference → <Addd 3 Line > → Ok → Ok



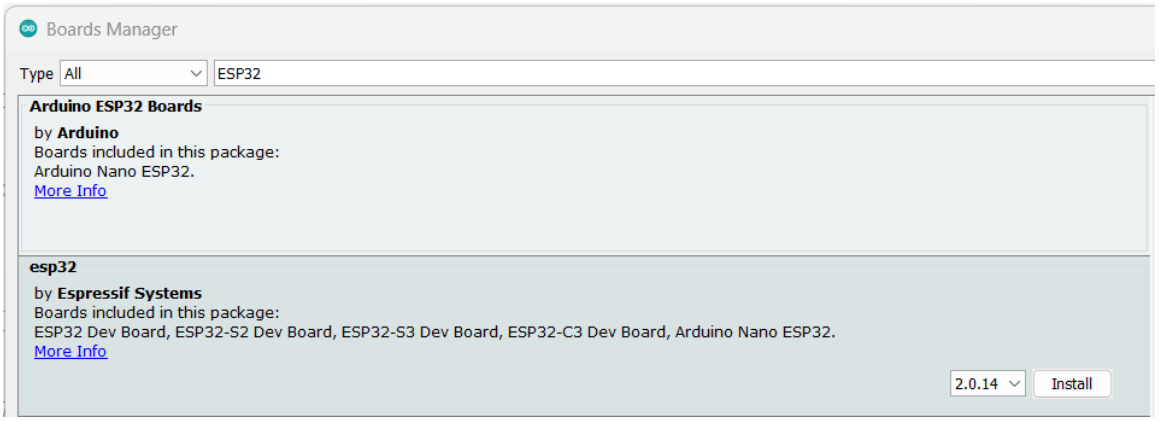

Additional Boards Manager URLs:

https://raw.githubusercontent.com/espressif/arduino-esp32/gh-pages/package_esp32_index.json

http://arduino.esp8266.com/stable/package_esp8266com_index.json

https://github.com/earlephilhower/arduino-pico/releases/download/global/package_rp2040_index.json

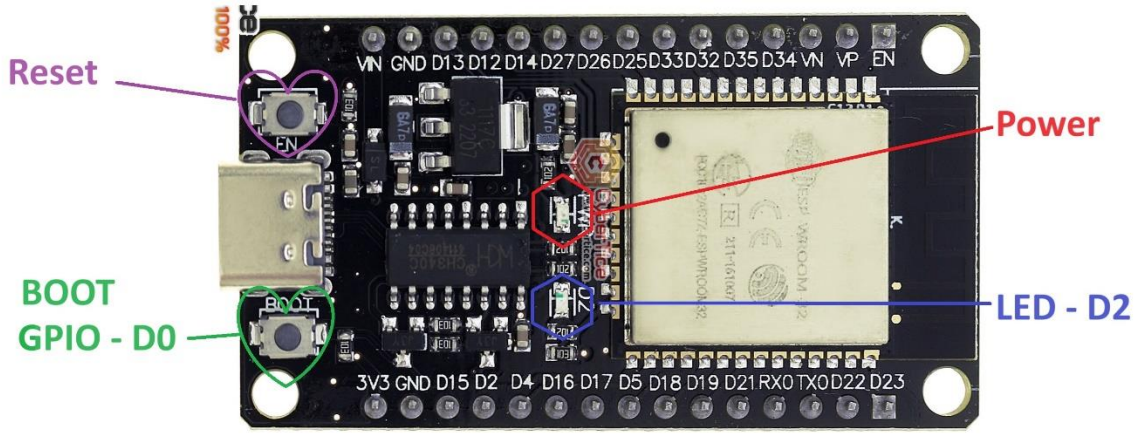
5. Add ESP32 Board: Tools → Boards → Board Manager




Filter = esp32

Select → esp32 by Espressif System → Version 2.0.14 → Install

6. Read https://www.allnewstep.com/b/144
7. Connect ESP32 Board to PC
8. Blink Test Code: File → Example → 01.Basic → Blink
9. Select Serial Port: Tools → Port → ComX
10. Select Target Board: Tools → Boards → ESP32 Arduino → DOIT ESP32 DEVKIT V1
11. Upload Code to Target Board: Sketch → Upload
12. ESP32 Pinout Reference https://randomnerdtutorials.com/esp32-pinout-reference-gpios/




Blink Test: On Board LED = LED_BUILTIN = GPIO D2

## 2/10. ESP32 GPIO

2.1 ESP32 - LED Output

2.2 ESP32 – Switch Input

13. Read https://www.allnewstep.com/b/157

## 3/10. ESP32 + Single Digit 7_Segment

## 4/10. ESP32 + Four Digit 7_Segment

## 5/10. ESP32 + Eight Digit 7_Segment with MAX_7219

## 6/10. ESP32 + Dot Matrix with MAX_7219

6.1 ESP32 + Dot Matrix

6.2 ESP32 + Dot Matrix with MAX_7219

## 7/10. ESP32 + TM_1638

## 8/10. ESP32 + DHT_22

## 9/10. ESP32 + Rotary Encoder

## 10/10. ESP32 + Ambient Sensor